DocuCare AI - Smart Healthcare Planning Assistant

A Project-II Report

Submitted in partial fulfillment of requirement of the

Degree of

BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE & ENGINEERING

BY

Animesh Bokil	**EN22CS301125**
Anshika Vyas	**EN22CS301154**
Aryan Singh	**EN22CS301215**
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Avani Rawal	**EN22CS301238**

Under the Guidance of

Dr. Hemlata Patel
Prof. Ajeet Singh Rajput
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MEDICAPS UNIVERSITY

Department of Computer Science & Engineering
Faculty of Engineering
MEDICAPS UNIVERSITY, INDORE- 453331

JAN-JUNE 2026

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JAN-JUNE 2026

Report Approval

The project work **"DocuCare AI – Smart Healthcare Planning Assistant"** is hereby approved as a creditable study of an engineering subject carried out and presented in a manner satisfactory to warrant its acceptance as prerequisite for the Degree for which it has been submitted.

It is to be understood that by this approval the undersigned do not endorse or approved any statement made, opinion expressed, or conclusion drawn there in; but approve the "Project Report" only for the purpose for which it has been submitted.

Internal Examiner

Name:

Designation

Affiliation

External Examiner

Name:

Designation

Affiliation

Declaration

We hereby declare that the project entitled **"DocuCare AI – Smart Healthcare Planning Assistant"** submitted in partial fulfillment for the award of the degree of Bachelor of Technology in 'B.Tech Computer Science' completed under the supervision of **Dr. Hemlata Patel, Prof. Ajeet Singh Rajput, Prof. Akshay Saxena,** Faculty of Engineering, Medicaps University Indore is an authentic work.

Further, We declare that the content of this Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for the award of any degree or diploma.

Signature and name of the student(s) with date

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Certificate

We, **Dr. Hemlata Patel, Prof. Ajeet Singh Rajput, Prof. Akshay Saxena** certify that the project entitled **"DocuCare AI – Smart Healthcare Content Assistant"** submitted in partial fulfillment for the award of the degree of Bachelor of Technology by **Animesh Bokil, Anshika Vyas, Aryan Singh, Avani Rawal** is the record carried out by them under our guidance and that the work has not formed the basis of award of any other degree elsewhere.

Prof. Hemlata Patel

Prof. Ajeet Singh Rajput

Prof. Akshay Saxena

Computer Science and Engineering

Medicaps University, Indore

Name of External Guide

Dr. Kailash Chandra Bandhu

Head of the Department

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We express our heartfelt gratitude to our Internal Guide**, Prof. Akshay Saxena, Prof. Ajeet Singh Rajput, Dr. Hemlata Patel, Professor, Department of Computer Science Engineering, Medicaps University, Indore** without whose continuous help and support, this project would ever have reached to the completion.

It is their help and support, due to which we became able to complete the design and technical report.

Without their support this report would not have been possible.

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Abstract

The healthcare industry involves complex documentation processes that require accurate management of patient information, clinical terminology, formatting standards, and consistency in communication. Traditional documentation methods are often manual, time-consuming, and prone to inconsistencies and errors. With the advancement of Artificial Intelligence, there is a growing need for intelligent systems that can automate and optimize healthcare content generation.

This project presents the development of an **Integrated Healthcare Content Generator Agent**, an AI-powered system designed to assist healthcare professionals in generating structured and high-quality medical documentation. The system utilizes an **agentic multi-agent architecture**, where a central intelligent planner processes user inputs and decomposes them into specific tasks such as content generation, medical validation, summarization, and formatting. These tasks are handled by specialized agents, and the results are aggregated to generate comprehensive outputs such as patient summaries and clinical reports.

The application is built using modern technologies including **Large Language Models (LLMs)** for understanding user queries and generating domain-specific content, along with **vector databases** for contextual knowledge retrieval and a backend system for orchestration and data handling. Advanced **prompt engineering techniques** are used to ensure consistency in tone, structure, and medical terminology. A user-friendly interface enables users to input healthcare-related data and receive structured outputs in real time.

Additionally, the system integrates **DevOps practices** such as Dockerization, CI/CD pipelines, and cloud deployment to ensure scalability, reliability, and continuous delivery of the application.

The proposed system improves documentation efficiency, reduces manual effort, and enhances decision-making by providing integrated and intelligent solutions. It demonstrates the potential of **GenAI and Agentic AI** in transforming traditional healthcare documentation into a more automated, scalable, and data-driven process.

This work highlights the practical application of AI-driven multi-agent systems in healthcare and their potential to revolutionize content generation processes by making them more intelligent, consistent, and efficient.

Keywords:

Artificial Intelligence, Generative AI, Agentic AI, Healthcare, LLM, Prompt Engineering, Vector Database, Medical Documentation, DevOps, Automation

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Abbreviations

The following abbreviations are used in the DocuCare AI – Smart Healthcare Planning Assistant project report:

- AI – Artificial Intelligence
- LLM – Large Language Model
- RAG – Retrieval-Augmented Generation
- API – Application Programming Interface
- UI – User Interface
- EC2 – Elastic Compute Cloud (AWS service)
- AWS – Amazon Web Services
- CI/CD – Continuous Integration and Continuous Deployment
- JSON – JavaScript Object Notation
- Docker – Containerization platform for application deployment
- Git – Version control system
- GitHub – Cloud-based Git repository hosting service
- ML – Machine Learning

Notations & Symbols

The following notations and symbols are used in the DocuCare AI project report for better understanding of system components and processes:

- AI / LLM – Artificial Intelligence / Large Language Model used for natural language understanding and planning.
- UI – User Interface
- API – Application Programming Interface used for communication between modules.
- EC2 – Amazon Elastic Compute Cloud used for deployment.
- CI/CD – Continuous Integration and Continuous Deployment pipeline for automation.
- Docker – Containerization platform used to package the application.
- Agent – Independent functional module in an Agentic AI system performing specific tasks (content generation, medical validation, summarization, and formatting).
- JSON – Data format used for structured communication between modules.

Chapter-1

Introduction

1.1 Introduction

The healthcare industry involves complex documentation processes that require coordination among clinical data, patient information, and standardized medical practices. Effective documentation is essential to ensure accurate patient records and proper communication among healthcare professionals. However, traditional documentation methods are often manual, time-consuming, and prone to human errors, leading to inefficiencies and increased workload.

With the advancement of Artificial Intelligence (AI) and Large Language Models (LLMs), there is significant potential to automate healthcare content generation. AI-driven systems can analyze medical inputs, process large volumes of data, and generate structured outputs. This reduces manual effort and improves accuracy and decision-making.

This project, **DocuCare AI – Smart Healthcare Planning Assistant** , aims to develop an intelligent system that generates structured healthcare content from user inputs. The system uses an agent-based approach to analyze requirements, break down tasks, and produce outputs such as patient summaries, clinical notes, and medical reports. This ensures systematic and adaptable content generation.

Furthermore, modern AI models can understand complex medical instructions and convert them into structured solutions. By integrating multiple components into a unified system, the proposed solution improves consistency, accuracy, and reduces variability in documentation.

Overall, the system focuses on automating healthcare content generation, improving efficiency, and minimizing human effort, while ensuring scalability and real-world applicability.

1.2 Literature Review

The field of healthcare documentation has evolved with the integration of digital technologies and Artificial Intelligence. Traditional methods of creating patient records are mostly manual and lack intelligent automation. Although electronic systems improve data storage, they still depend on manual input and do not provide advanced content generation capabilities.

Recent advancements in Artificial Intelligence and Large Language Models (LLMs) have enabled automation in healthcare content generation. These systems can process clinical inputs and generate structured medical content. Agent-based approaches further improve efficiency by organizing tasks into smaller components.

AI tools are increasingly used to automate medical documentation and analysis. However, challenges such as maintaining accuracy, ensuring consistency in medical terminology, and handling real-world scenarios still remain. This project addresses these challenges by integrating GenAI and agent-based architecture to develop a more reliable and efficient healthcare content generation system.

1.3 Objectives

The main objectives of the DocuCare AI – Smart Healthcare Planning Assistant project are:

- To develop an AI-based system for automating healthcare content generation
- To convert user inputs into structured and professional medical documents
- To reduce manual effort and minimize errors in healthcare documentation
- To improve efficiency, consistency, and accuracy of clinical content
- To implement an agent-based approach for task decomposition and execution
- To generate scalable and adaptable planning solutions for different healthcare documentation needs.

1.4 Significance of the Study

This project is significant as it addresses major challenges in traditional healthcare documentation methods.

- Helps in reducing time and effort required for medical documentation
- Improves decision-making by providing structured and AI-generated insights
- Provides automation in healthcare content generation processes
- Ensures consistency and accuracy in medical terminology and documentation
- Demonstrates the application of GenAI and multi-agent systems in real-world healthcare scenarios

Overall, the system contributes to the development of intelligent and automated healthcare documentation solutions.

1.5 Research Design

- Based on an **applied research and system development approach**
- Focuses on designing and implementing an AI-based healthcare content generation system
- **Problem Identification:** Analyze limitations of traditional healthcare documentation.
- **Requirement Analysis:** Identify user needs, inputs, and system functionalities
- **System Design:** Plan architecture including frontend, backend, and AI agent modules
- **Development:** Build the system using suitable programming tools and frameworks
- **AI Integration:** Implement agent-based logic for task decomposition and planning
- **Testing:** Evaluate system performance, accuracy, and usability
- **Deployment:** Use Docker and CI/CD pipelines for scalable and automated deployment
- **Evaluation:** Analyze results to validate effectiveness of the proposed system

1.6 Source of Data

The data used in this project is derived from multiple sources to support the development and testing of the system. These include:

- **User Inputs:** Clinical information, symptoms, notes, and healthcare-related details.
- **Simulated Data:** Artificial datasets created to test different healthcare documentations.
- **System-Generated Data:** Outputs produced by the AI agent during planning and execution

The combination of these data sources ensures that the system is flexible and capable of handling various types of healthcare content generation tasks. Proper preprocessing and structuring of data are performed to maintain consistency, accuracy, and reliability.

S. No	Data Source	Description
1	User Input	Healthcare inputs like symptoms and notes provided by user
2	LLM Models	AI-generated medical content outputs
3	Predefined Rules	Medical formatting and validation rules
4	System Logs	Intermediate agent processing outputs

Table 1.1

1.7 Chapter Scheme

The project report is organized into the following chapters to provide a structured and systematic presentation of the work:

Chapter 1: Introduction

This chapter presents an overview of the project, including the background, literature review, objectives, significance, research design, source of data, and chapter scheme. It provides the foundation for understanding the need for the DocuCare AI – Smart Healthcare Planning Assistant and defines the scope of the system.

Chapter 2: Present Investigation

This chapter describes the experimental setup and the procedures adopted for developing and testing the DocuCare AI – Smart Healthcare Planning Assistant. It explains how the system was implemented and evaluated in a structured environment.

Chapter 3: System Design

This chapter explains the overall system design and architecture, including application design, process flow, information flow, and component design. It describes how different modules such as frontend, backend, and AI agents interact with each other.

Chapter 4: Implementation

This chapter focuses on the development details of the system, including the development environment, technology stack, module-wise implementation, and API integration used in the DocuCare AI – Smart Healthcare Content Generator .

Chapter 5: DevOps and Deployment

This chapter explains the DevOps practices used in the project, including Dockerization, CI/CD pipeline, and deployment on AWS EC2. It describes how the system is made scalable and production-ready.

Chapter 6: Results and Discussions

This chapter presents the system output and discusses the performance of the DocuCare AI – Smart Healthcare Planning Assistant. It also highlights the challenges faced during development and how they were resolved.

Chapter 7: Summary and Conclusions

This chapter provides a summary of the complete project and presents the final conclusions drawn from the implementation and results of the system.

Chapter 8: Future Scope

This chapter describes possible future enhancements of the DocuCare AI – Smart Healthcare Planning Assistant, including improvements in AI models, scalability, real-time integration, and advanced features.

Chapter-2

Present Investigation

2.1 Experimental Set-up

The experimental setup for the proposed system is designed to develop and evaluate an AI-based healthcare content generation system. The system is implemented using a combination of frontend and backend technologies along with AI models for intelligent content generation. The backend is responsible for processing user inputs, managing logic, and interacting with the AI agents. The frontend provides a user-friendly interface where users can input healthcare-related data and view generated outputs.The system is containerized using Docker to ensure portability and consistency across different environments. CI/CD pipelines are used to automate the building, testing, and deployment processes. The setup also includes testing environments where different healthcare documentation scenarios are simulated to evaluate system performance.

Component	Details
Operating System	Windows / Linux
Programming Language	Python 3.x
Frameworks	FastAPI/Flask
AI Model	Large Language Models (LLM)
Version Control	Git & GitHub
Deployment	AWS EC2

Table 2.1

2.2 Procedures Adopted

The following procedures are adopted for the development and evaluation of the system:

- Collecting and analyzing user requirements for healthcare content generation.
- Designing the system architecture including frontend, backend, and AI modules
- Developing the application using appropriate programming tools and frameworks
- Integrating the AI agent to process inputs and generate structured plans
- Testing the system with different healthcare scenarios and inputs
- Evaluating the accuracy, consistency, and usability of the generated content
- Deploying the system using Docker and CI/CD pipelines

Chapter 3

System Design

3.1 Application Design

The application follows a multi-layered agent-based architecture where user inputs are processed through an AI planner, decomposed into tasks, and executed via specialized modules.

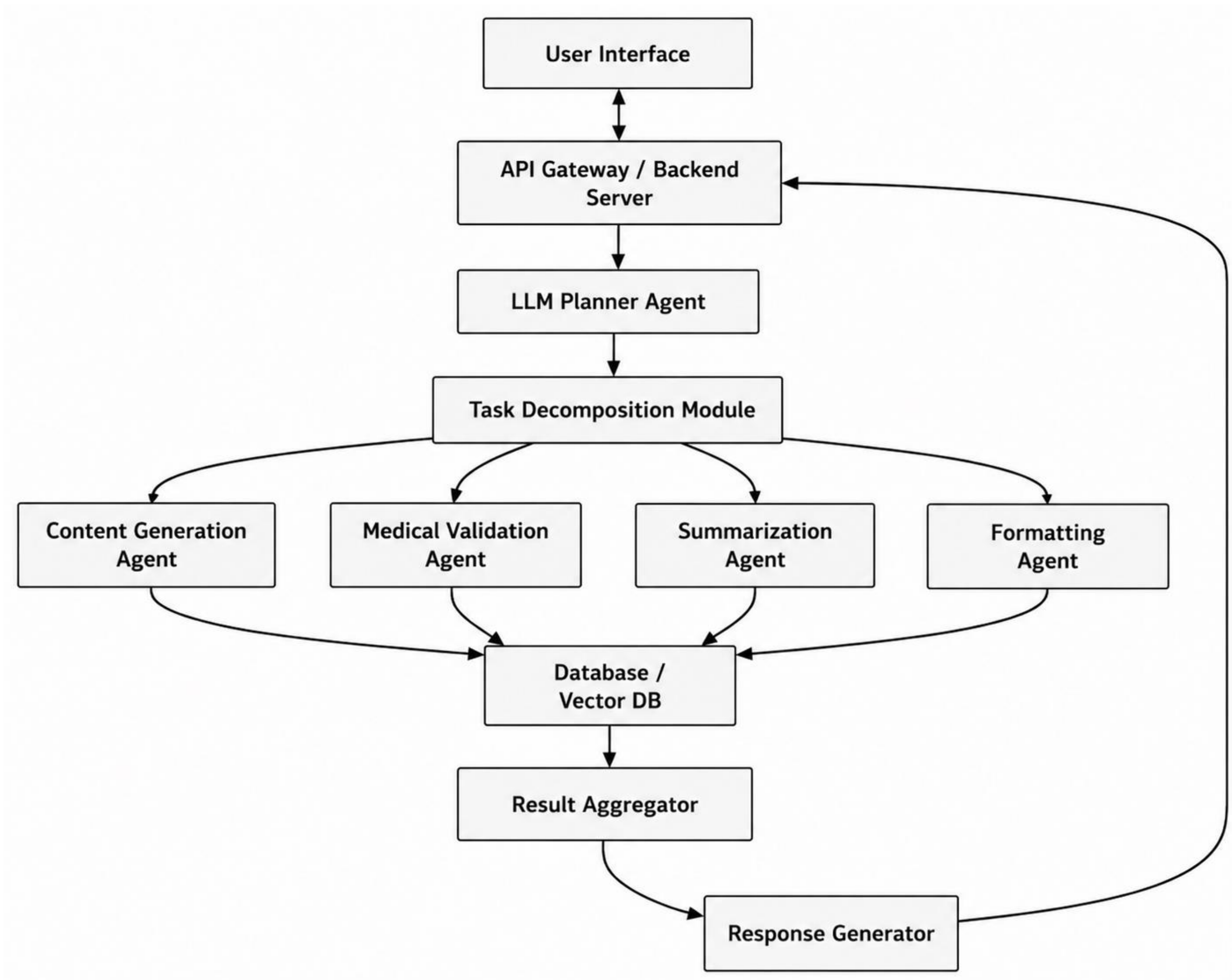


Figure 3.1

3.2 Process Flow

This defines the **step-by-step execution pipeline** from user input to final output.

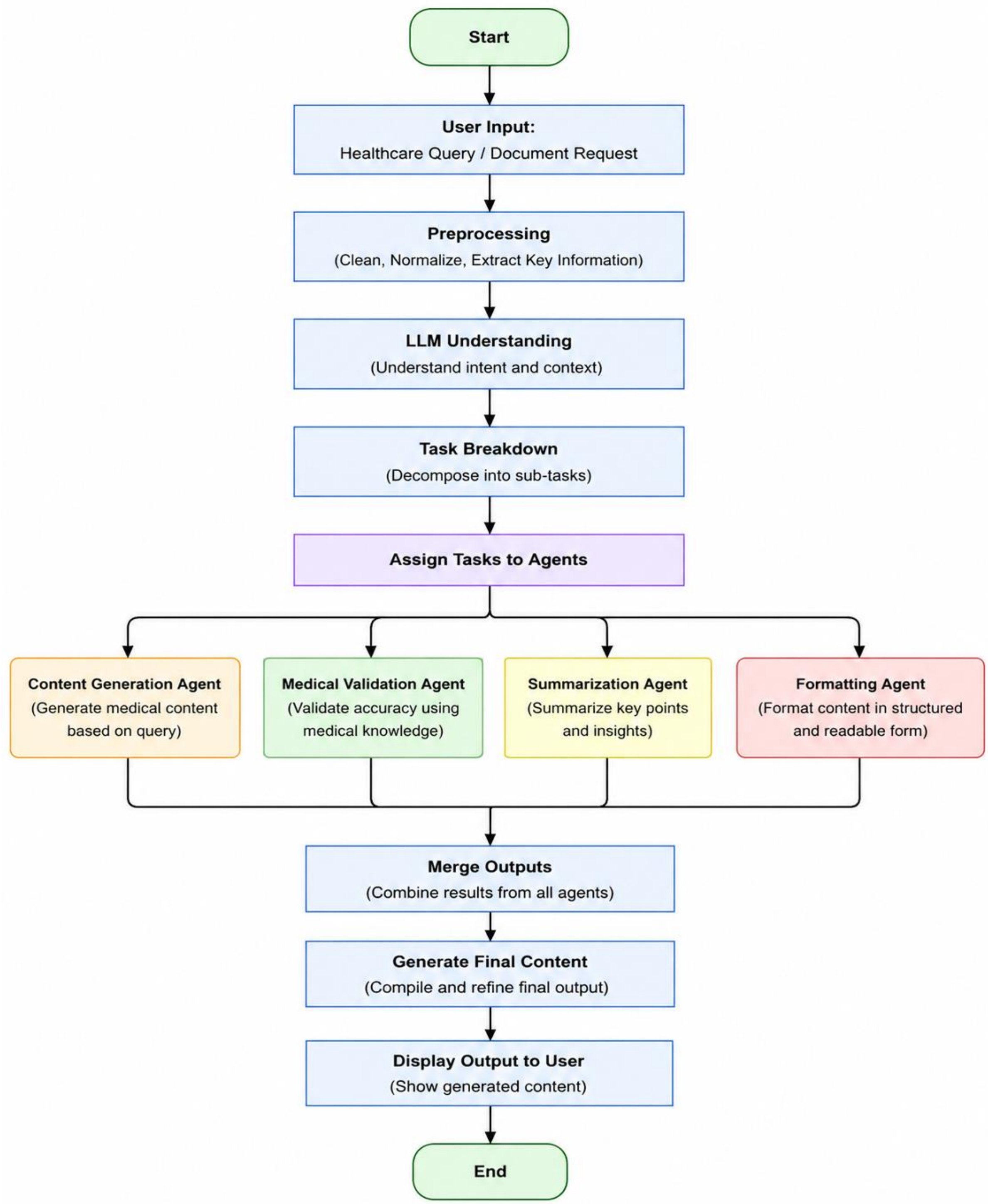


Figure 3.2

3.3 Information Flow

This shows how data moves across components including inputs, intermediate outputs, and final results.

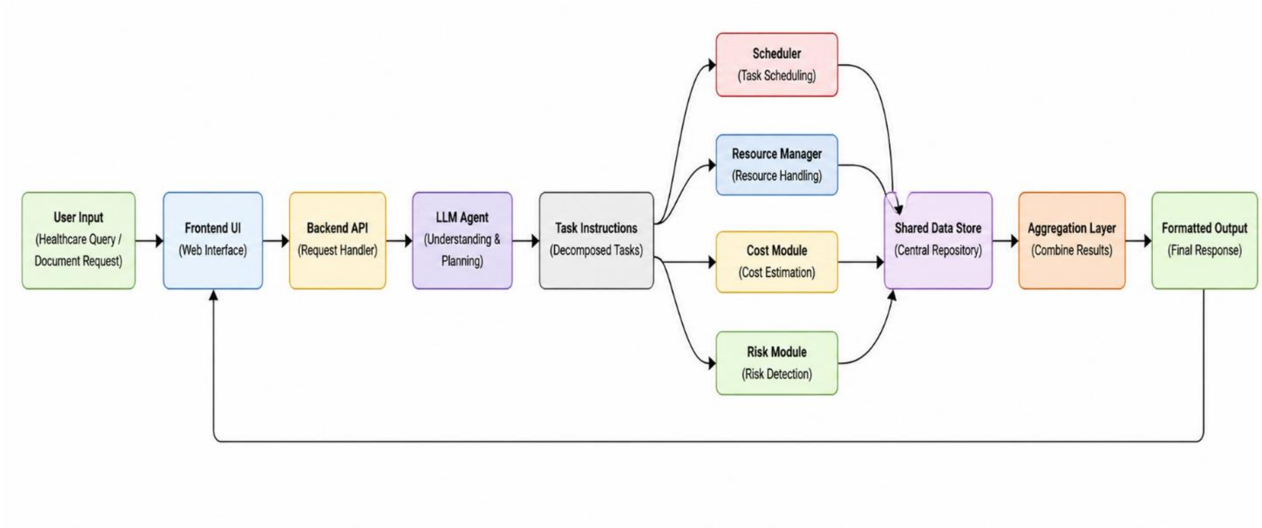


Figure 3.3

3.4 Components Design

Each component is modular and loosely coupled to ensure scalability and maintainability.

Key Components

- Frontend (HTML,CSS,JS) – User interaction
- Backend API (FastAPI/Flask) – Handles requests
- LLM Planner Agent – Core intelligence (task understanding)
- Task Decomposer – Breaks query into subtasks
- Domain Agents:
    - Content Generation Agent
    - Medical Validation Agent
    - Summarization Agent
    - Formatting Agent
- Database – Stores project data
- Aggregator – Combines outputs
- Response Generator – Final output formatting

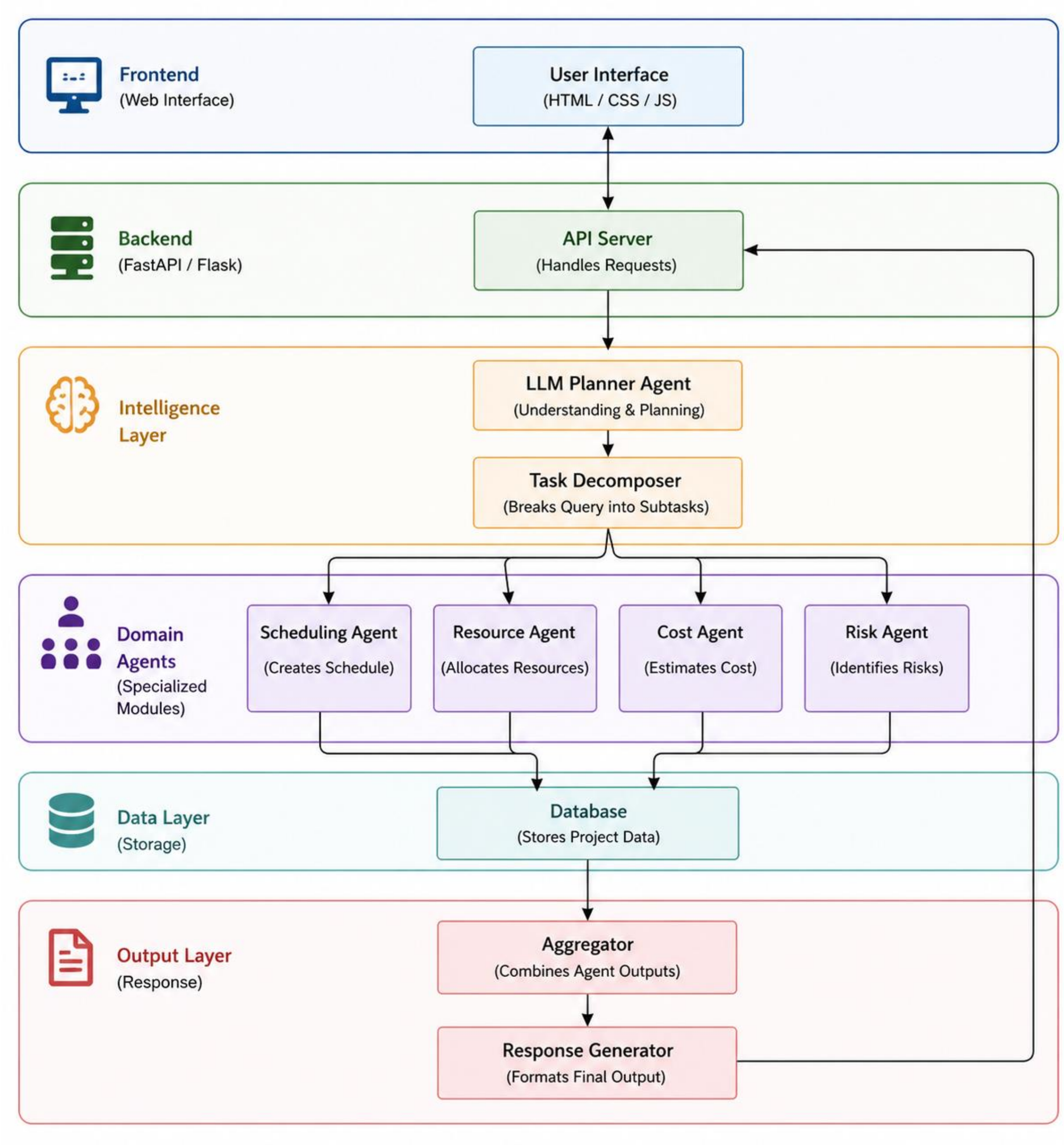

Frontend
(Web Interface)
User Interface
(HTML / CSS / JS)
Backend
(FastAPI / Flask)
API Server
(Handles Requests)
Intelligence Layer
LLM Planner Agent
(Understanding & Planning)
Task Decomposer
(Breaks Query into Subtasks)
Domain Agents
(Specialized Modules)
Scheduling Agent
(Creates Schedule)
Resource Agent
(Allocates Resources)
Cost Agent
(Estimates Cost)
Risk Agent
(Identifies Risks)
Data Layer
(Storage)
Database
(Stores Project Data)
Output Layer
(Response)
Aggregator
(Combines Agent Outputs)
Response Generator
(Formats Final Output)

Figure 3.4

Chapter 4

Implementation

4.1 Development Environment

The development of the DocuCare AI – Smart Healthcare Planning Assistant was carried out in a structured and efficient environment to ensure smooth integration of all components.

- Operating System: Windows / Linux
- Development Tools: Visual Studio Code (VS Code)
- Programming Language: Python (due to strong AI/ML ecosystem)
- Version Control: Git and GitHub (for code management and collaboration)
- Environment Management: Virtual Environment (venv) for dependency isolation
- Runtime: Python 3.x

The system was tested locally and can be extended to cloud or container-based environments for deployment.

4.2 Technology Stack

The system is developed using modern tools and frameworks:

- Programming Language: Python
- Frontend: HTML,CSS,JS (for interactive UI)
- Backend: FastAPI / Flask (API handling and orchestration)
- AI Engine: Large Language Models (LLMs)
- Agent Framework: Multi-agent orchestration (CrewAI / custom agents)
- Database: JSON / lightweight storage (or optional DB)
- Deployment Tools: Docker, Kubernetes (optional based on setup)

4.3 Module-wise Implementation

The DocuCare AI – Smart Healthcare Planning Assistant is implemented in a modular way to ensure clarity, scalability, and easy maintenance. Each module is responsible for a specific function within the system.

1. User Interface Module

This module is developed using HTML,CSS,JS and provides an interactive interface for users. It accepts natural language input related to Healthcare planning and displays the generated outputs such as patient summaries, clinical notes, and structured medical content in a structured format.

2. Input Processing Module

This module preprocesses the user input by cleaning and structuring the text. It ensures that the input is properly formatted before being sent to the LLM-based planner for further processing.

3. LLM Planner Module

This is the core intelligence module of the system. It interprets the user query using a Large Language Model and converts it into structured healthcare content generation tasks. It identifies key requirements such as symptoms, medical context, and relevant clinical information.

4. Task Decomposition Module

This module breaks the main healthcare content generation task into smaller sub-tasks. These sub-tasks are then assigned to specialized agents for execution.

5. Agent Modules

The system includes multiple specialized agents:

- **Content Generation Agent** for generating healthcare-related content.
- **Medical Validation Agent** for ensuring accuracy and correctness of medical information
- **Summarization Agent** for extracting key points from the generated content
- **Formatting Agent** for structuring the content into a clear and readable format

Each agent processes its assigned task independently.

6. Aggregation Module

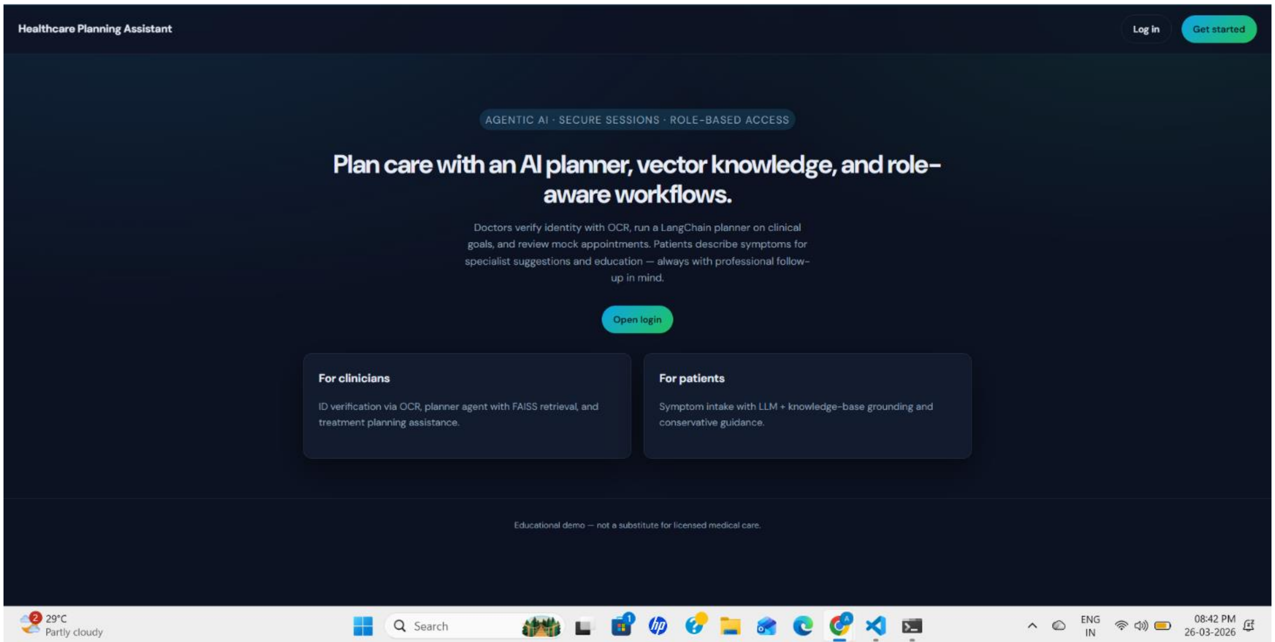
This module collects outputs from all agents and combines them into a unified and structured healthcare document.

7. Output Generation Module

The final module formats the aggregated results and presents them in a user-friendly manner through the interface.

4.4 API Integration

The DocuCare AI – Smart Healthcare Planning Assistant uses API integration to ensure smooth communication between the frontend interface, backend services, and AI-based modules. The system follows a REST-based architecture where the user input from the frontend interface is sent to the backend through API requests. The backend processes these requests and forwards the data to the LLM Healthcare Planner Module, which generates structured healthcare plans based on the input. These plans are then distributed to different agent modules such as scheduling, resource allocation, cost estimation, and risk analysis. Each module processes its assigned task and returns the results to the backend through internal API calls.



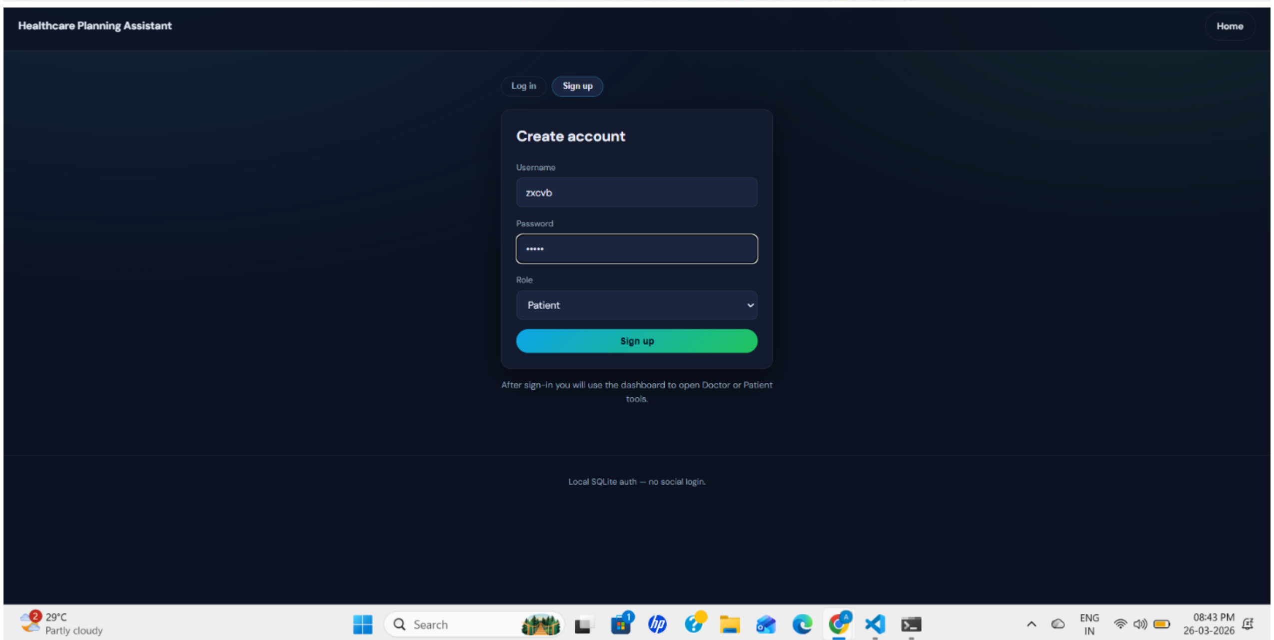
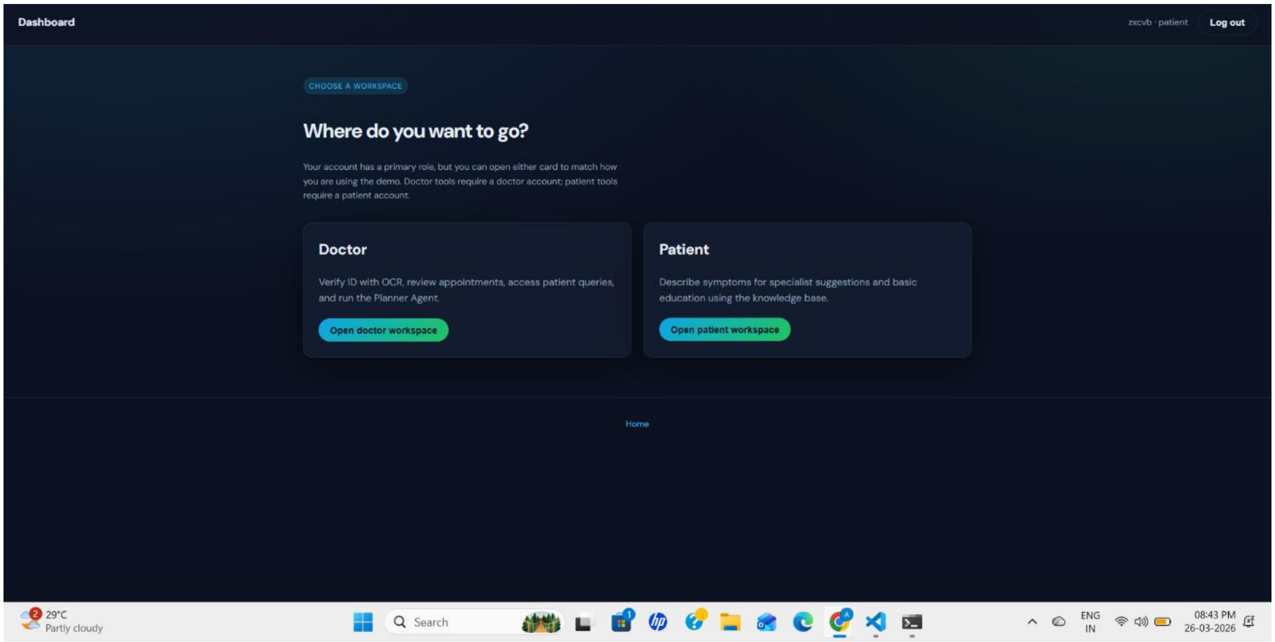


Figure 4.1, Figure 4.2



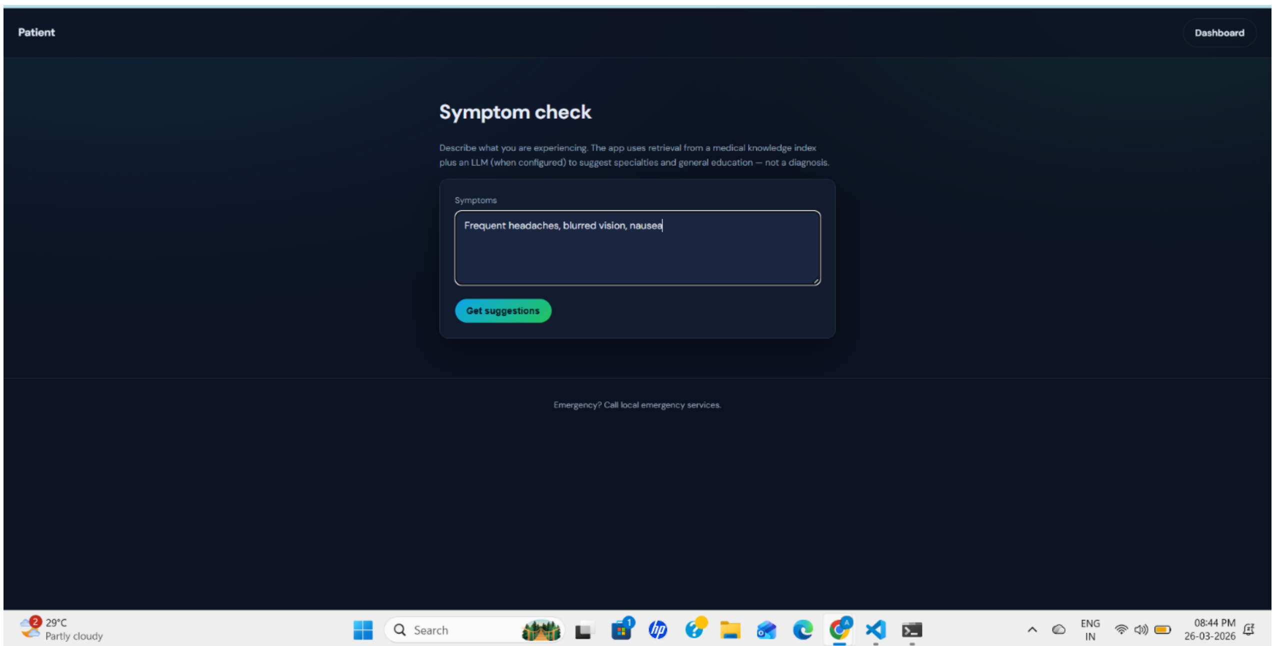


Figure 4.3, Figure 4.4

Chapter 5

Devops and Deployment

5.1 Dockerization

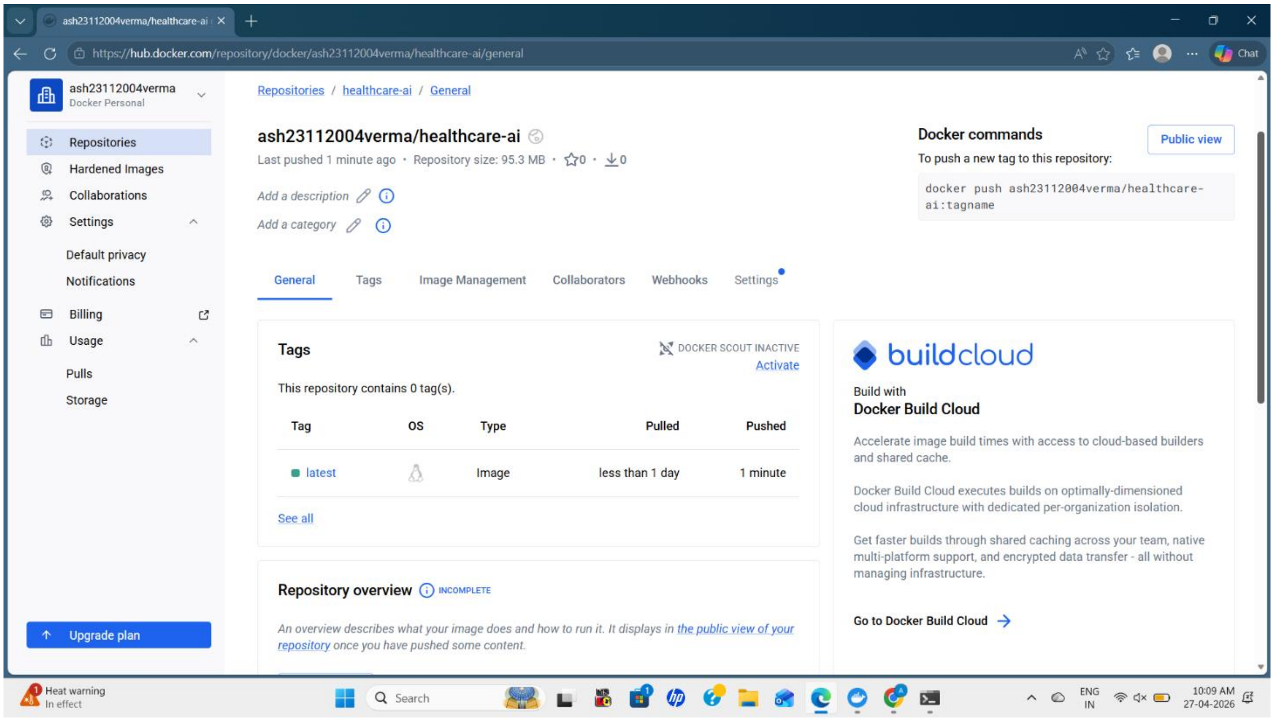


Figure 5.1

5.2 CI/CD Pipeline

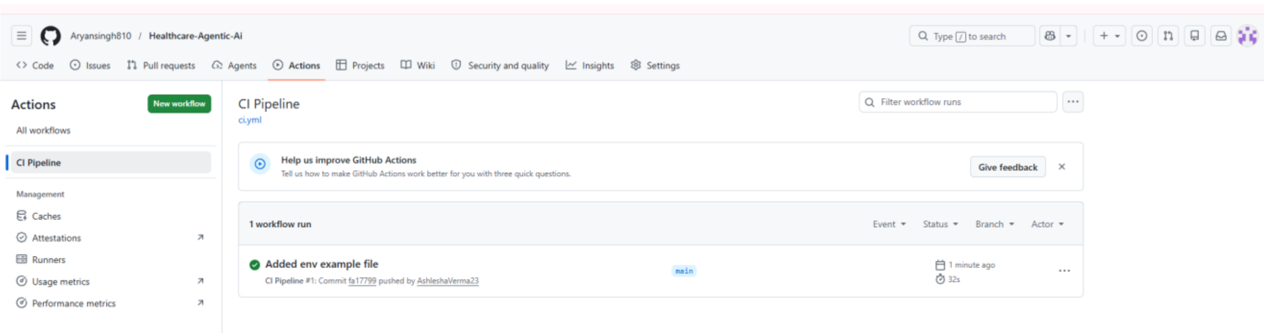


Figure 5.2

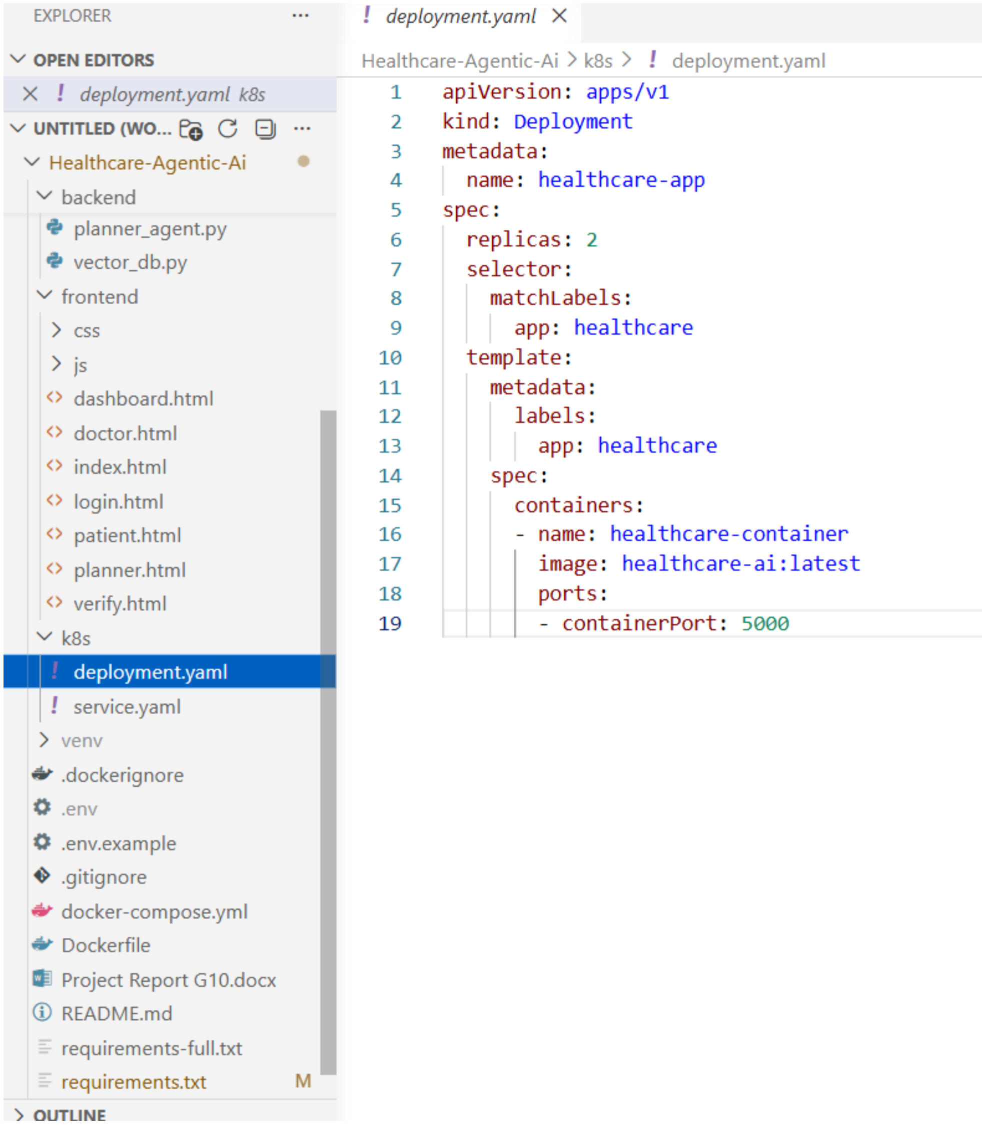

EXPLORER
OPEN EDITORS
deployment.yaml k8s
UNTITLED (WO...
Healthcare-Agentic-Ai
backend
planner_agent.py
vector_db.py
frontend
css
js
dashboard.html
doctor.html
index.html
login.html
patient.html
planner.html
verify.html
k8s
deployment.yaml
service.yaml
venv
.dockerignore
.env
.env.example
.gitignore
docker-compose.yml
Dockerfile
Project Report G10.docx
README.md
requirements-full.txt
requirements.txt
M
OUTLINE
deployment.yaml
Healthcare-Agentic-Ai > k8s > deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
name: healthcare-app
spec:
replicas: 2
selector:
matchLabels:
app: healthcare
template:
metadata:
labels:
app: healthcare
spec:
containers:
- name: healthcare-container
image: healthcare-ai:latest
ports:
- containerPort: 5000

Figure 5.3

5.3 AWS EC2 Deployment

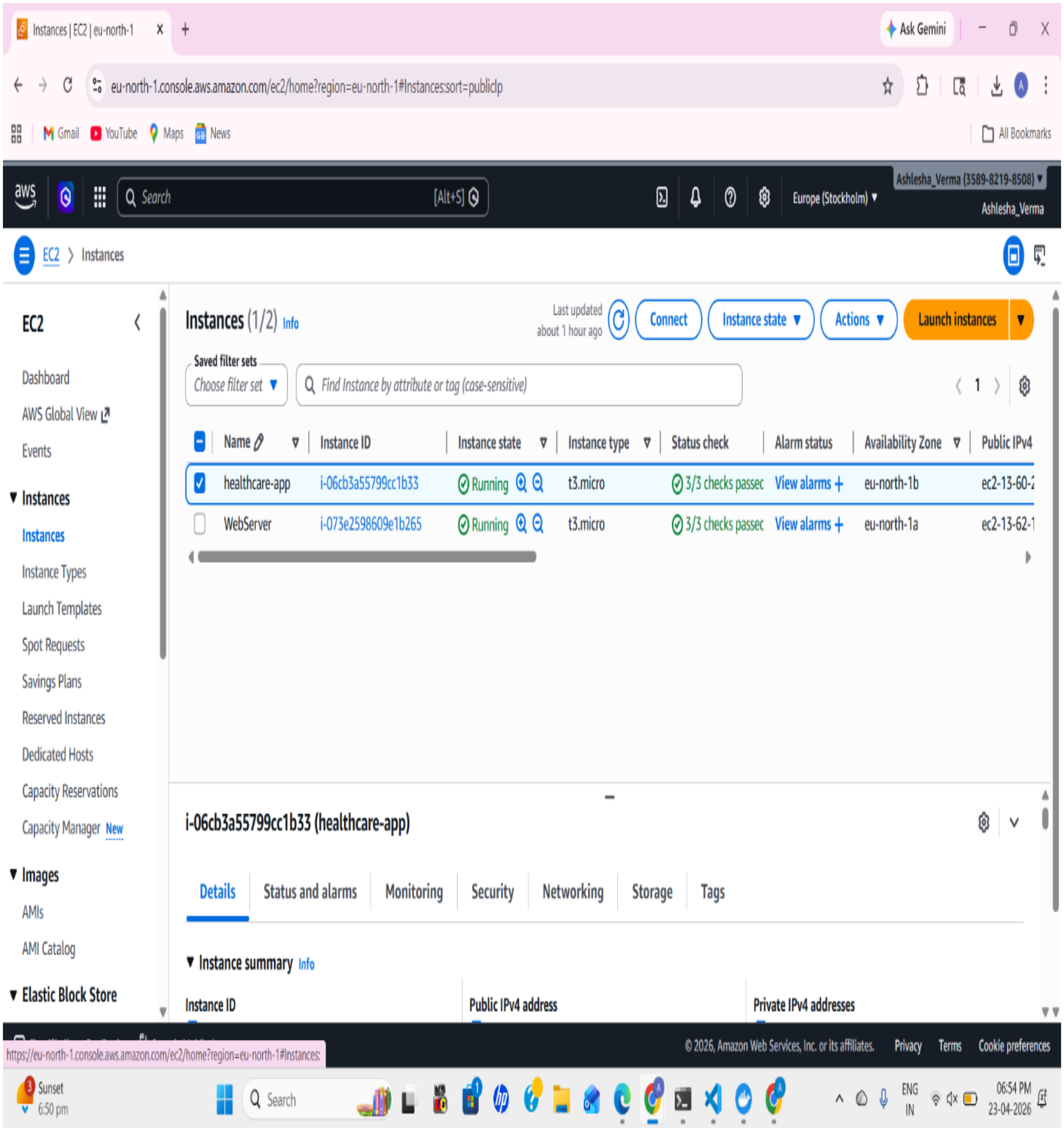


Figure5.4

Chapter 6

Results and Discussions

6.1 System Output

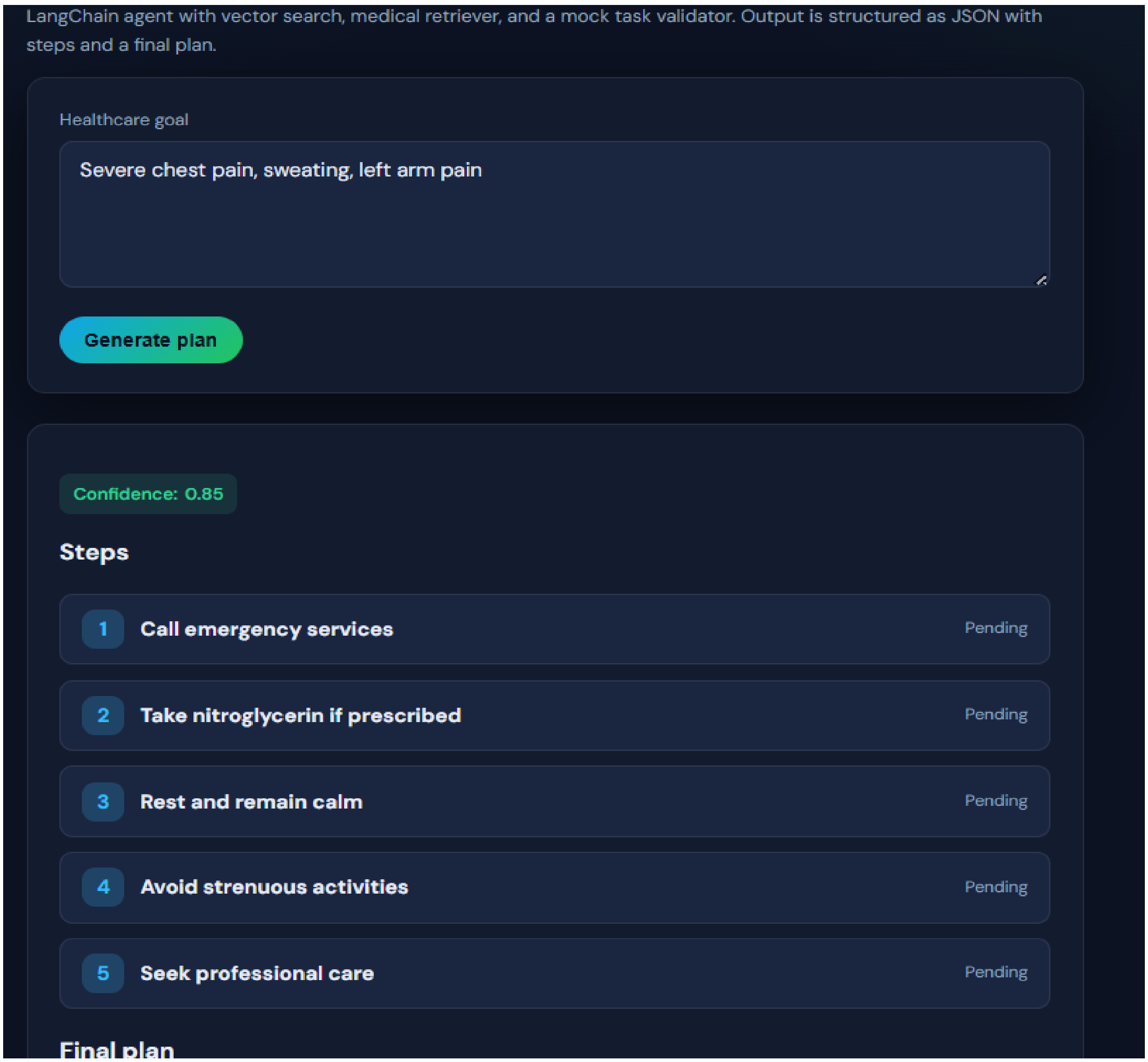


Figure 6.1

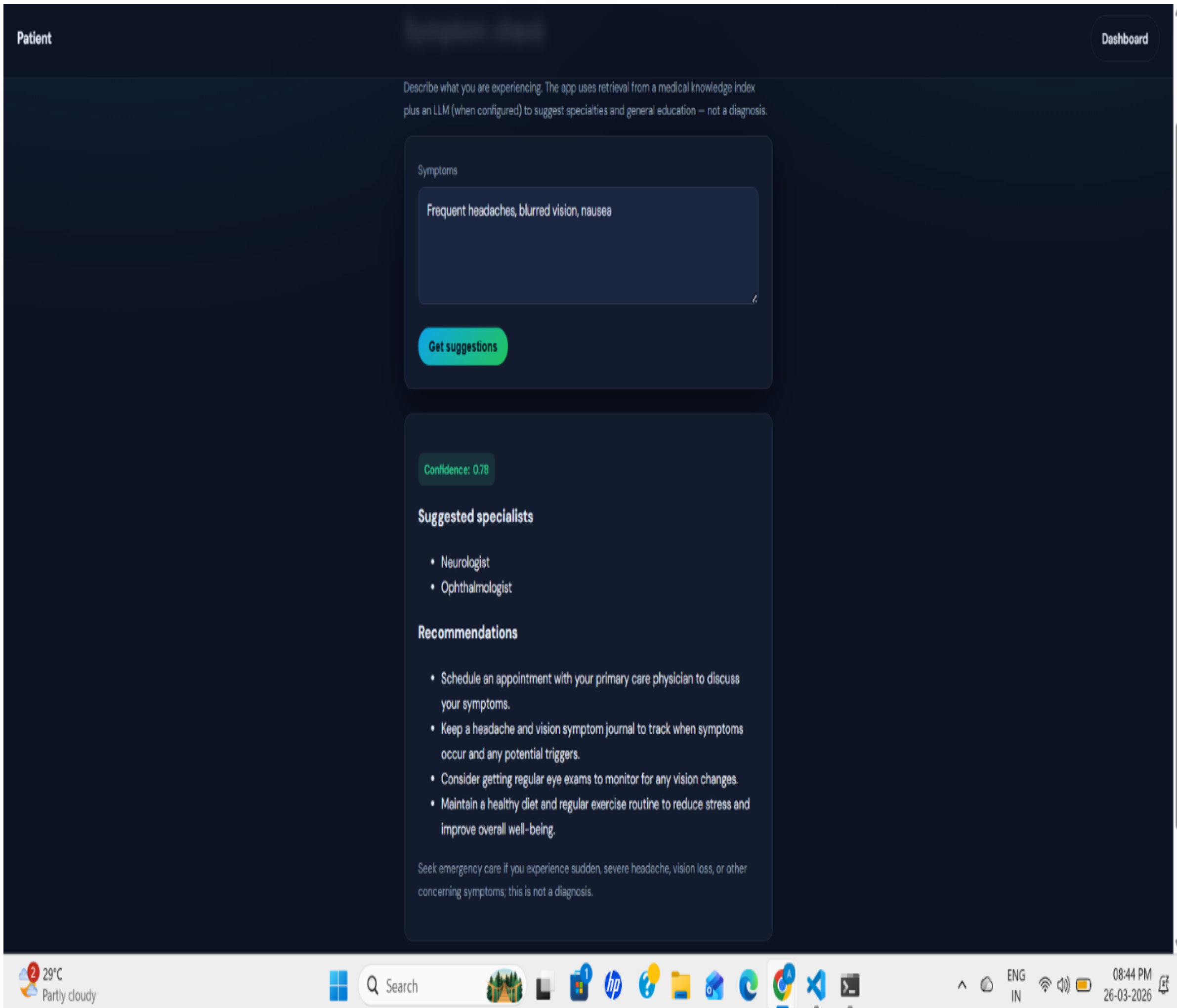

Patient
Dashboard
Describe what you are experiencing. The app uses retrieval from a medical knowledge index plus an LLM (when configured) to suggest specialties and general education — not a diagnosis.
Symptoms
Frequent headaches, blurred vision, nausea
Get suggestions
Confidence: 0.78
Suggested specialists
Neurologist
Ophthalmologist
Recommendations
Schedule an appointment with your primary care physician to discuss your symptoms.
Keep a headache and vision symptom journal to track when symptoms occur and any potential triggers.
Consider getting regular eye exams to monitor for any vision changes.
Maintain a healthy diet and regular exercise routine to reduce stress and improve overall well-being.
Seek emergency care if you experience sudden, severe headache, vision loss, or other concerning symptoms; this is not a diagnosis.
29°C
Partly cloudy
Search
ENG
IN
08:44 PM
26-03-2026

Figure 6.2

6.2 Challenges Faced

During the development of the **DocuCare AI – Smart Healthcare Content Generator**, several technical and integration challenges were encountered. One of the key challenges was coordinating multiple AI agents, as each module (such as content generation, medical validation, summarization, and formatting) needed to function independently while still producing consistent and unified healthcare outputs. Ensuring seamless communication between the frontend, backend, and LLM modules was also challenging due to complex API integrations and continuous data exchange.

Another major challenge was optimizing system performance, as LLM processing and multiple agent interactions introduced latency and affected response time. Additionally, maintaining accuracy, consistency, and context-awareness in healthcare content required careful handling of dynamic user inputs. These challenges were addressed through a modular and scalable system design, well-structured API communication, and continuous testing and refinement, resulting in a reliable, efficient, and adaptable system.

Chapter 7

Summary and Conclusions

7.1 Summary

The **DocuCare AI – Smart Healthcare Content Generator** is an AI-based system designed to automate and improve healthcare documentation. The system integrates Large Language Models with a multi-agent architecture to handle key tasks such as content generation, medical validation, summarization, and formatting. The application provides a user-friendly interface through which users can input healthcare-related data in natural language and receive structured and professional medical outputs. The system is implemented using modular components, ensuring scalability, flexibility, and easy maintenance. It is also deployed on AWS EC2 with Docker and CI/CD support for efficient and reliable operation.

The system integrates a multi-agent architecture with Large Language Models to interpret user inputs and generate structured healthcare content. It provides a simple and interactive interface that allows users to describe clinical information in natural language. The backend processes these inputs and coordinates between different agents to produce complete and accurate healthcare documents. The application is designed using a modular approach and deployed on AWS EC2, ensuring scalability, accessibility, and reliable performance.

7.2 Conclusion

The project successfully demonstrates the application of Artificial Intelligence and multi-agent systems in the field of healthcare documentation. The **DocuCare AI – Smart Healthcare Planning Assistant** reduces manual effort, improves accuracy, and enhances efficiency by automating complex documentation processes. The integration of Large Language Models and specialized agents enables intelligent task decomposition and structured content generation. The cloud-based deployment ensures accessibility and scalability of the system. Overall, the project highlights how modern AI technologies can transform traditional healthcare documentation into a more automated, consistent, and efficient process.

The project demonstrates an effective use of Artificial Intelligence in improving healthcare content generation. The system reduces dependency on manual documentation and enhances efficiency through automated processing and intelligent decision-making. The integration of LLMs and multiple specialized agents ensures accurate and structured outputs for healthcare scenarios. The system is scalable, modular, and deployable in cloud environments, making it suitable for real-world applications. Overall, the project successfully achieves its objective of building an intelligent assistant that modernizes and optimizes healthcare documentation.

Chapter 8

Future Scope

Future Scope

- Integration of advanced LLMs for improved accuracy in healthcare content generation
- Development of a mobile application for easy access and real-time usage
- Integration with Electronic Health Records (EHR) systems for real-time data usage.
- Implementation of advanced analytics for better healthcare insights
- Support for multi-user collaboration in clinical environments
- Addition of chatbot and voice-enabled interfaces for better user interaction

Appendix

The Appendix section provides additional supporting information related to the **DocuCare AI – Smart Healthcare Planning Assistant** project that helps in better understanding of the system implementation and usage.

Appendix A: Project Structure

```
docucare-ai-healthcare-content-generator/
│
├── app.py                          # Streamlit UI
├── main.py                         # Backend  entry point
├── requirements.txt                # Dependencies
├── agents/
│   ├── content_generation_agent.py    # Generates healthcare content
│   ├── medical_validation_agent.py    # Validates medical accuracy
│   ├── summarization_agent.py         # Summarizes key information
│   └── formatting_agent.py            # Formats content for output
├── utils/
│   ├── helper_functions.py            # Utility and helper functions
│   └── prompt_templates.py            # Prompt templates for agents
├── data/
│   ├── sample_inputs.json             # Sample healthcare inputs
│   └── medical_knowledge.txt          # Reference medical knowledge
├── templates/
│   └── report_template.html           # HTML template for formatted output
├── static/
│   ├── style.css                      # CSS for UI styling
│   └── logo.png                       # Application logo
└── Dockerfile                      # Containerization file
```

Figure 8.1

Appendix B: Sample Input

- "Generate a patient summary based on symptoms including fever, cough, and fatigue."

Appendix C: Sample Output

- Patient summary with symptoms and clinical observations
- Structured medical content (diagnosis overview and recommendations)
- Key point summary of patient condition
- Formatted healthcare report with clear sections.

Appendix D: Tools Used

- Python 3.x
- FastAPI / Flask
- Docker
- AWS EC2
- Git & GitHub
- Large Language Models (LLMs)

Appendix E: System Workflow Overview

User Input → UI (HTML, CSS, JS) → Backend API → LLM Planner → Agents (Content Generation, Medical Validation, Summarization, Formatting) → Aggregation → Final Output

Bibliography

The following references and resources were used for the development and understanding of the Construction Planning Assistant Agent project:

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4. FastAPI Documentation, https://fastapi.tiangolo.com
5. Groq Documentation, *"Groq API and Developer Guide"*, https://console.groq.com/docs
6. Docker Documentation, https://docs.docker.com
7. AWS EC2 Documentation, https://docs.aws.amazon.com/ec2
8. GitHub Repository: https://github.com/Aryansingh810/Healthcare-Agentic-Ai.git
9. CrewAI / Multi-Agent System Concepts (General Reference on AI Agent Architecture)
10. Research papers on AI in Healthcare and Medical Documentation (IEEE / Springer publications)